Q1. (20 *pts.*) Consider a CT LTI system with input $x(t)$ and output $y(t)$ described by the following differential equation:

$$x(t-\pi) + \frac{dx(t)}{dt} + 2\int_{-\infty}^{t} x(\tau)\,d\tau = y(t)$$

a) By utilizing the CT Fourier transform properties, find the frequency response $H(\omega)$ of the system for a signal with $X(0) = 0$.
b) If $x_1(t) = e^{jt}$, what is the corresponding output $y_1(t)$?
c) If $x_2(t) = e^{jt} + e^{3jt}$, what is the corresponding output $y_2(t)$?
d) Is $e^{jt} + e^{3jt}$ an eigenfunction of this system?

a) → take F.T of both sides

$$e^{-j\pi w} \cdot X(w) + jw\,X(w) + 2 \cdot X(w)/_{jw} + 2 \cdot \pi \cancel{X(0)}^{0} \cdot \delta(w) = Y(w)$$

$$X(w)\left(2/_{jw} + jw + e^{-j\pi w}\right) = Y(w) \rightarrow \boxed{H(w) = \frac{Y(w)}{X(w)} = 2/_{jw} + jw + e^{-j\pi w}}$$

b) $x_1(t) = e^{jt}$ → eigen function

↳ $y_1(t) = x_1(t)\,H(w_0)$ → $w_0 = 1$ $\quad -j^2 - j$

↳ $y_1(t) = e^{jt} \cdot \left(2/_j + j + e^{-j\pi}\right) = e^{jt} \cdot (-j - 1)$

$$y_1(t) = (\cos t + j\sin t) = (-j-1)\cos t + (1-j)\sin t)$$

c) $x_2(t) = e^{jt} + e^{3jt}$ ⇒ sum of eigen function.

$$y_2(t) = e^{jt} \cdot H(1) + e^{3jt} \cdot H(3) = e^{jt}\left(2/_j + je^{-j\pi}\right) + e^{3jt}\left(2/_{3j} + 3j + e^{-j3\pi}\right)$$

$$y_2(t) = e^{jt}(-j-1) + e^{3jt}\left(7/_3 j - 1\right)$$

d) $e^{jt} + e^{3jt}$ is not eigenfunction of this system but is sum of two eigen functions of this system.

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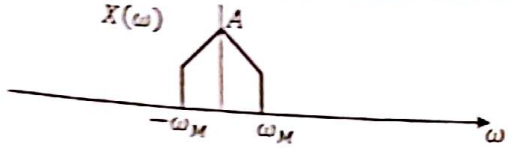
$f_s/2\pi < 1/2\omega_m$

Q2. (20 pts.) Consider the band-limited signal $x(t)$ with real and even spectrum $X(\omega)$ plotted below.



a) Determine the range of sampling frequencies ω_s and sampling periods T_s that will allow perfect recovery of $x(t)$ from its samples.

This is a low pass signal. For a low pass signal our condition is $\omega_s > 2\omega_m$ for perfect recovery

Nyquist Rate $\boxed{\omega_s > 2\omega_m}$

$\omega_s = 2\pi/T_s \rightarrow 2\pi/T_s > 2\omega_m \rightarrow \boxed{T_s < \pi/\omega_m}$

Assume for the remaining parts of the question (part-b and part-c), the sampling frequency ω_s is chosen according to the Nyquist sampling theorem.

b) Suppose $x(t)$ is sampled using an impulse train as shown in the figure below. Derive the spectrum $X_p(\omega)$ in terms of $X(\omega)$ and also plot $X_p(\omega)$.

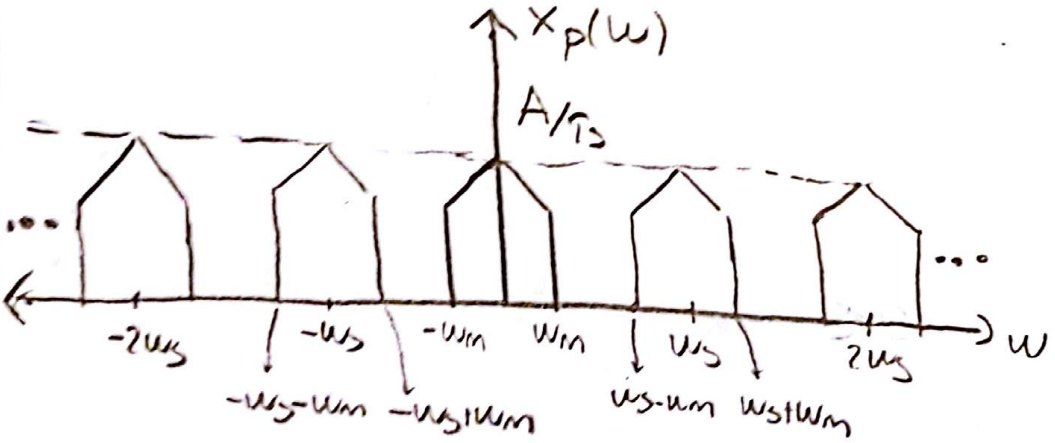
$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

$1/T_s$

$x(t) \cdot p(t) = x_p(t) \longleftrightarrow X_p(\omega) = 1/2\pi\, X(\omega) * P(\omega)$

$P(\omega) = 2\pi/T_s \sum_{k=-\infty}^{\infty} \delta(\omega - k \cdot 2\pi/T_s) \rightarrow X_p(\omega) = 1/T_s \left(\sum_{k=-\infty}^{\infty} \delta(\omega - k2\pi/T_s) \right) * (X(\omega))$

$\boxed{X_p(\omega) = 1/T_s \sum_{k=-\infty}^{\infty} X(\omega - k.\omega_s)} \rightarrow \omega_s = 2\pi/T_s$



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$\text{rect}(t/2\tau_1) \longleftrightarrow 2\tau_1 \,\text{sic}(\omega\tau_1/\pi)$ $\quad \frac{2\omega_1}{\pi}\text{sinc}\left(\frac{t\omega_1}{\pi}\right) \longleftrightarrow \text{rect}(\omega/2\omega_1)$

c) Suppose $x_p(t)$ from part-b is passed through an LTI system with impulse response $h_a(t)$ and frequency response $H_a(\omega)$. If it is desired that the output $x_a(t)$ is an amplitude modulated version of $x(t)$ with carrier frequency ω_s, i.e., $x_a(t) = x(t)\cos(\omega_s t)$, determine mathematical expressions for both $H_a(\omega)$ and $h_a(t)$. Plot also $H_a(\omega)$.

$$x_p(t) \longrightarrow \boxed{h_a(t)} \longrightarrow x_a(t) = x(t)\cos(\omega_s t)$$

$$X_a(t) = x(t)\cos(\omega_s t) \longleftrightarrow X_a(\omega) = \frac{1}{2\pi} X(\omega) * C(\omega)$$

$$C(\omega) = F\{\cos(\omega_s t)\} = \pi \cdot (\delta(\omega-\omega_s) + \delta(\omega+\omega_s))$$

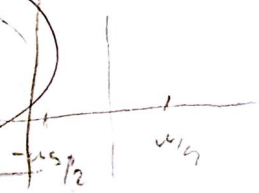
$$X_a(\omega) = \frac{1}{2}(\delta(\omega-\omega_s) + \delta(\omega+\omega_s)) * X(\omega)$$

$$\boxed{X_a(\omega) = \frac{1}{2}(X(\omega-\omega_s) + X(\omega+\omega_s))}$$

$\rightarrow$ Gain of $H_a(\omega)$ $\rightarrow$ $\frac{A}{T_s} \cdot G = \frac{1}{2}$ $\rightarrow$ $\boxed{G = T_s/2A}$ $\rightarrow$

$H_a(\omega)$ has to be a band pass filter and centered around ω_s and $-\omega_s$.

$$H_a(\omega) = \begin{cases} T_s/2A, & \omega_s/2 \le |\omega| \le 3\omega_s/2 \\ 0, & \text{ow} \end{cases}$$



$$\boxed{H_a(\omega) = T_s/2A \left(\text{rect}(\omega/\omega_s)\right) * (\delta(\omega-\omega_s) + \delta(\omega+\omega_s))}$$ ok.

$$(\delta(\omega-\omega_s) + \delta(\omega+\omega_s)) \longleftrightarrow \frac{1}{\pi}\cos(\omega_s t)$$

$$\text{rect}(\omega/\omega_s) \longleftrightarrow \frac{\omega_s}{2\pi}\text{sinc}\left(\frac{t\omega_s}{2\pi}\right)$$

$$H_a(\omega) \longleftrightarrow T_s/2A \cdot 2\pi \cdot \frac{1}{\pi}\cos(\omega_s t) \cdot \frac{\omega_s}{2\pi} \cdot \text{sinc}(t\omega_s/2\pi) = h_a(t)$$

$$\boxed{h_a(t) = \frac{T_s \omega_s}{\pi \cdot 2A}\cos(\omega_s t)\sin(t\omega_s/2\pi)}$$

$\frac{7}{8}$

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Q3. (20 pts.)

Consider a signal $x(t)$ whose spectrum is as follows:

$$X(\omega) = \begin{cases} e^{-2\omega}, & \text{if } \omega \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

a) Find $x(t)$.
b) Find the total energy in the signal $x(t)$.

Consider also an LTI system with impulse response $h(t) = \frac{1}{\pi}\,\text{sinc}\left(\frac{t+2}{\pi}\right)$.

c) Find the frequency response of the system.
d) Let $y(t)$ be the output when $x(t)$ above is input to this system. Find $Y(\omega)$, i.e., the spectrum of the output. Also plot the magnitude and phase of $Y(\omega)$.

a) $x(t) = \frac{1}{2\pi}\int_0^\infty e^{-2\omega} \cdot e^{-j\omega t}\, d\omega = \frac{1}{2\pi} \cdot \left(-\frac{1}{2+jt} e^{-\omega(2+jt)}\right)\Big|_0^\infty$

$= \frac{1}{2\pi} \cdot \frac{1}{2+jt}$

$\boxed{x(t) = \frac{1}{2\pi}(2+jt)}$

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b) $\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$: Parseval Relation

$= \frac{1}{2\pi}\int_0^\infty e^{-4\omega} d\omega = \frac{1}{2\pi} \cdot \left(-\frac{1}{4} e^{-4\omega}\right)\Big|_0^\infty = \boxed{\frac{1}{8\pi}}$ → total energy in $x(t)$.

c) $\frac{1}{\pi}\,\text{sinc}\left(\frac{t}{\pi}\right) \longleftrightarrow \text{rect}\left(\frac{\omega}{2}\right)$

$\frac{1}{\pi}\,\text{sinc}\left(\frac{t+2}{\pi}\right) \longleftrightarrow e^{j2\omega}\,\text{rect}\left(\frac{\omega}{2}\right)$ → time shifting property

$\boxed{H(\omega) = e^{j2\omega}\,\text{rect}\left(\frac{\omega}{2}\right)}$

→

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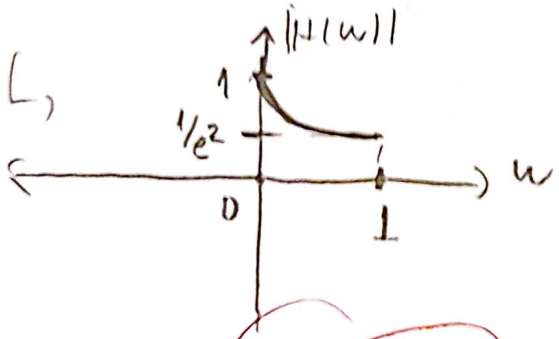
d) $y(t) = x(t) * h(t) \longleftrightarrow Y(w) = X(w)H(w)$ → convolution property

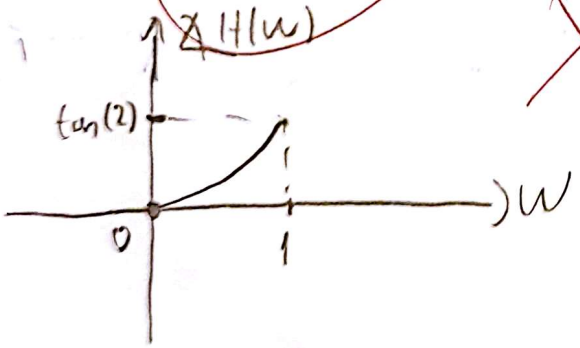
$$Y(w) = X(w)\, e^{j2w} \operatorname{rect}(w/2) = \begin{cases} e^{j2w-2w} & , 0 \le w \le 1 \\ 0 & , \text{otherwise} \end{cases}$$

- $|H(w)| = 1/e^{2w}$,



- $\measuredangle H(w) = \tan(2w)$

sin(0)



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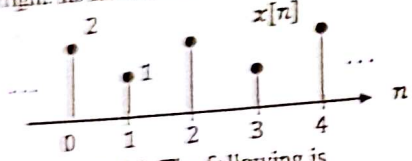
Q4. (20 pts) (All parts can be solved independently.)

a) The signal $x[n]$ is the periodic signal shown on the right. Its fundamental period is 2.

Using DTFS, express $x[n]$ as a linear combination of complex exponentials. Also find the DTFS coefficients a_k.



b) Another signal $y[n]$ is a periodic signal with fundamental period 3. The following is known about its DTFS coefficients b_k:

- $b_1 = j$
- $b_0 = 2$
- $b_{-1} = -j$

$\frac{2\pi}{2}$

Find the signal $y[n]$. Simplify your result so that it only contains sinusoids and constants.

c) Now consider $z[n] = x[n] + y[n]$. Verify that $z[n]$ is periodic. Also find its fundamental period.

d) Let c_k be the DTFS coefficients of $z[n]$. Find c_k for k=0, 1,...,5 in terms of the DTFS coefficients of $x[n]$ and $y[n]$, i.e., in terms of a_k and b_k. Comment on your result.

a)

$$a_k = \frac{1}{2}\sum_{n=0}^{1} x[n]\, e^{-j2\pi kn} \quad \Rightarrow \Omega_0 = \pi$$

$$a_k = \frac{1}{2}\left(x[0]\cdot 1 + x[1]\, e^{-j\pi k}\right) = \frac{1 + e^{-j\pi k}}{2} \quad \Rightarrow \boxed{a_k = \frac{1+e^{-j\pi k}}{2}}$$

and $\boxed{x[n] = \sum_{k=\langle 2\rangle} a_k \cdot e^{j\pi kn}}$ ok!

b) b_k is periodic with $N=3$

$\hookrightarrow b_1 = j,\ b_0 = b_3 = 2,\ b_{-1} = -j$

$$y[n] = \sum_{k=1}^{1} b_k\, e^{j\Omega_0 kn} = -j\cdot e^{-j2\pi/3\, n} + 2 + j\, e^{j2\pi/3\, n}$$

$\hookrightarrow \Omega_0 = 2\pi/3$

$$= j\cdot\left(e^{j2\pi/3\, n} - e^{-j2\pi/3\, n}\right) + 2$$

$$= j\cdot 2j\cdot \sin(2\pi/3\, n) + 2 = -2\sin(2\pi/3\, n) + 2$$

$$\boxed{y[n] = -2\sin(2\pi/3\, n) + 2}$$

$\rightarrow$

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c) let say $N = 2.3 = 6$

$z[n+6] = x[n+2.3] + y[n+2.3]$

$\Rightarrow x[n+2.3] = x[n]$ → it's periodic with 2.

$y[n+2.3] = y[n]$ → it's periodic with 3.

then $z[n+6] = x[n] + y[n] = z[n]$ ✓

↳ $z[n]$ is periodic with $N_0 = 6$. ✓

↳ fundamental period.

d)

$n/3 = n'$ $0 - 2$

$$C_k = 1/6 \sum_{n=0}^{5} z[n] \cdot e^{-j\pi/3 kn} = 1/6 \underbrace{\sum_{n=0}^{5} x[n] e^{-j\pi/3 kn}}_{n/3 = n'} + 1/6 \underbrace{\sum_{n=0}^{5} y[n] e^{-j\pi/3 kn}}_{n/2 = n'}$$

$$= \underbrace{1/6 \sum_{n'=0}^{1} x[3n'] e^{-j\pi kn'}}_{1/6\, a_k} + \underbrace{1/6 \sum_{n'=0}^{2} y[2n'] e^{-j2\pi/3 kn'}}_{1/6\, b_k}$$

$$\boxed{C_k = 1/6 (a_k + b_k)}$$

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Q5. (20 *pts.*) The impulse response of an LTI system is $\delta[n] - 2\delta[n-1] + \delta[n-2]$.

a) Find the frequency response of the system and sketch its magnitude spectrum in the interval $[-\pi, \pi]$.

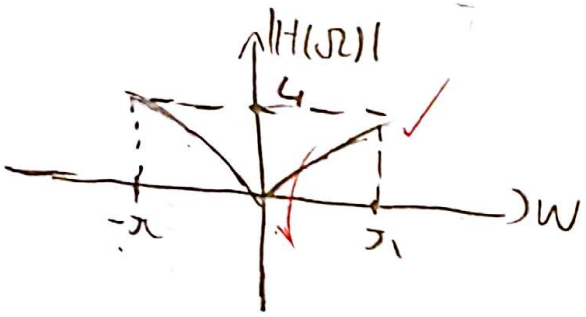
b) Determine the output signal $y[n]$ in compact form if the system is excited by the input signal $x[n] = \sin\left(\frac{\pi}{2}n\right) + (-1)^n + 2$. Specify the filter type (i.e., lowpass, bandpass or highpass) by comparing the input and output signals.

c) Find the Fourier Transform of $x[n]$ and plot its magnitude for $\Omega \in [-2\pi, 2\pi]$.

a) $H(\Omega) = 1 - 2.e^{-j\Omega} + e^{-j2\Omega}$: $\rightarrow H(\Omega)$ is periodic with 2π,

$|H(-\pi)| = |1 - 2e^{j\pi} + e^{j2\pi}| = 4$ $\quad |H(0)| = 0$

$|H(\pi)| = |1 - 2e^{-j\pi} + e^{-j2\pi}| = 4$



$1 - 2.(-j) + 1$

$1 - 2j - 1$

$1 + 2 + 1$

b) $x[n] = \frac{1}{2j}\left(e^{j\pi/2\, n} - e^{-j\pi/2\, n}\right) + e^{-j\pi n} + 2e^{0}$

$y[n] = \frac{1}{2j}\left(e^{j\pi/2\, n}.H(\pi/2) - e^{-j\pi/2\, n}.H(-\pi/2)\right) + e^{-j\pi n}.H(-\pi) + 2H(0)$

$= \frac{1}{2j}\left(2j.e^{j\pi/2\, n} - e^{-j\pi/2\, n}.(-2j)\right) + e^{-j\pi n}.4 + 2.0$

$y[n] = e^{j\pi/2\, n} + e^{-j\pi/2\, n} + 4.e^{-j\pi n} = 2.\cos(\pi/2\, n) + 4.(-1)^n$

* We see a constant term 2 in input whose spectrum is centered around zero. But it couldn't pass through system. And also sin function and $(-1)^n$ could pass through system whose one of them has a spectrum centered at $\mp\pi/2$, and other has a spectrum centered around $\mp\pi$. For this reason, system is high pass filter.

$\longrightarrow$

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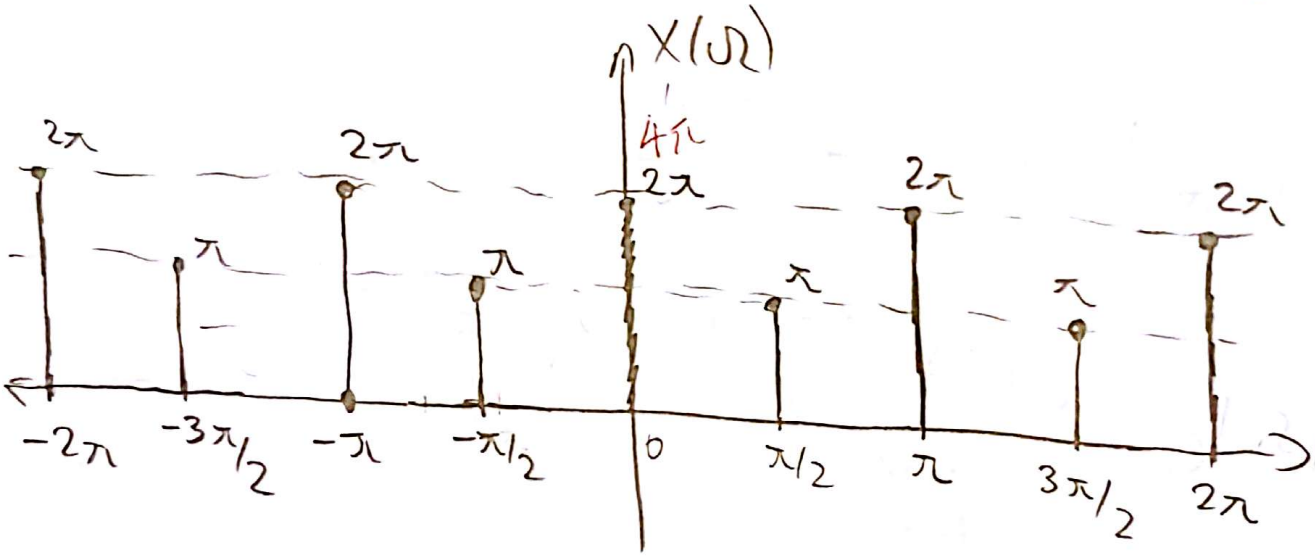
c)

$$\sin\left(\frac{\pi}{2}n\right) \longleftrightarrow \sum_{l=-\infty}^{\infty} \frac{\pi}{j}\left(\delta\left(\Omega-\frac{\pi}{2}-2\pi l\right)-\delta\left(\Omega+\frac{\pi}{2}-2\pi l\right)\right)$$

$$(-1)^n = e^{-j\pi n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \cdot \delta(\Omega+\pi-2\pi l)$$

$$2 \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi\,\delta(\Omega-2\pi l)$$

$$X(\Omega) = \sum_{l=-\infty}^{\infty}\Big[2\pi\,\delta(\Omega-2\pi l)+2\pi\,\delta(\Omega+\pi-2\pi l) + \frac{\pi}{j}\left(\delta\left(\Omega-\frac{\pi}{2}-2\pi l\right)-\delta\left(\Omega+\frac{\pi}{2}-2\pi l\right)\right)\Big]$$



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